

**Inclusive Educational Software System: Emotionally Expressive
Reading for Children with
Disabilities (Blind, Deaf, Cognitive Impairments)**

Project ID: 25-26J-407

Project Proposal Report

Brinthapan.J

B.Sc. (Hons) Degree in Information Technology Specializing in Information
Technology

Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

**Inclusive Educational Software System: Emotionally Expressive
Reading for Children with
Disabilities (Blind, Deaf, Cognitive Impairments)
Emotionally expressive text to sign language and user interaction**

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ABSTRACT

With the growing emphasis on inclusive education, the need for innovative tools to support children with sensory challenges has become increasingly critical. Deaf and hard-of-hearing children often face barriers in accessing written content, particularly in understanding emotional contexts embedded in text. This project aims to address these challenges by developing an **Emotionally Expressive Reader** that translates written text into sign language, enriched with emotional expressions. The primary goal is to create an interactive, engaging, and accessible system that not only converts text into sign language but also conveys the emotional tone, thereby enhancing comprehension and storytelling experiences for children.

The core of this study involves designing a **text-to-sign translation engine** integrated with natural language processing (NLP), sentiment analysis, and an avatar-based sign language interpreter. NLP ensures accurate mapping of text into sign language glosses, while sentiment analysis identifies emotional cues and integrates them into the avatar's facial expressions and gestures. The system will be deployed through a child-friendly mobile/tablet application, providing real-time translation, replay options, and interactive features to improve accessibility.

This research will investigate the existing challenges in text-to-sign translation, such as limited expressiveness, lack of emotional context, and usability gaps in current systems. By leveraging **AI-driven sentiment analysis and avatar-based animation**, the proposed solution will address these challenges, contributing to more inclusive and engaging digital learning tools for children with sensory needs. This study forms part of a broader initiative to enhance accessibility and inclusivity in educational technologies.

Keywords: Text-to-Sign Language, Deaf Education, Emotionally Expressive Reader, Natural Language Processing, Sentiment Analysis, Sign Language Animation, Inclusive Learning

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DECLARATION

I hereby declare that the work presented in this report of proposal is my own and has not been submitted elsewhere. All sources of information have been duly acknowledged. This report is submitted in partial fulfillment of the requirements of the above degree program, and it reflects the research and development plan for my individual component of the project.

Name	Student ID	Signature
Brinthapan.J	IT22123954	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



.....

Signature of the Supervisor

Ms. Sanjeevi Chandrasiri

.....27.08.2025.....

Date



.....

Signature of the Co-Supervisor
Ms . Karthiga Rajendran

.....27.08.2025.....

Date

LIST OF ABBREVIATIONS

Keyword	Meaning
IOT	Internet Of Things
UML	Unified Modeling Language
NLP	natural language processing
TTS	text-to-sign translation

1. INTRODUCTION

Children with sensory challenges, particularly those who are deaf or hard of hearing, often encounter significant barriers in accessing educational resources in ways that are both meaningful and engaging. Traditional reading tools rely heavily on text or audio, which may not fully address the communication needs of these learners. More importantly, such tools often lack the ability to convey the **emotional context** of content—an essential element for comprehension, empathy, and expressive language development.

While some digital solutions, such as text-to-speech or basic sign language animation tools, have been introduced, they typically focus only on functional delivery of content and overlook the **expressive and emotional dimension of communication**. This limitation can lead to reduced engagement, diminished comprehension, and weaker emotional connection to learning materials.

Recent advances in **artificial intelligence, natural language processing, and animated avatar technology** have opened new opportunities to create more inclusive and emotionally intelligent educational tools. By integrating emotion recognition and expressive sign language animation, it becomes possible to design a system that not only translates text into sign language but also embeds **facial expressions, gestures, and emotional tone**, thereby making communication more natural and impactful.

This research proposes the development of an **Emotionally Expressive Reader** that converts written text into expressive sign language animations for children with sensory challenges. Unlike earlier systems that emphasize basic accessibility, this project focuses on enhancing **emotional engagement, comprehension, and inclusivity** in education. The proposed solution aims to support children in better understanding learning materials, while also fostering stronger connections between content, emotion, and communication.

In doing so, this project contributes to the broader vision of inclusive education technologies that empower children with sensory challenges to learn, grow, and thrive in an environment that respects both their linguistic and emotional needs.

1.1 Background & Literature Survey

Inclusive education is one of the most critical challenges in today's society, especially for children with sensory difficulties such as hearing impairments. Traditional educational methods rely heavily on text and spoken language, making it difficult for children with sensory challenges to fully grasp learning content. While sign language serves as a primary mode of communication for many deaf and hard-of-hearing learners, existing digital learning platforms often fail to incorporate expressive and emotional elements that are crucial for effective communication.

Reading is not only about decoding text but also about understanding the **emotional tone, intent, and context** behind it. However, most existing text-to-sign language tools lack emotional depth, focusing only on the literal translation of words. This often results in limited comprehension, reduced engagement, and a diminished learning experience.

Advancements in **artificial intelligence (AI), natural language processing (NLP), computer vision, and 3D animation** now provide opportunities to design systems that can integrate emotion recognition and expressive sign language animations. Such systems can help children not only understand words but also feel the **emotional meaning** behind them, thereby creating a more engaging and supportive learning environment.

This research aims to build upon existing work by developing an **Emotionally Expressive Reader** that transforms written text into expressive sign language animations with integrated facial cues and emotional tones. Unlike traditional systems that prioritize accessibility alone, this project emphasizes **emotional engagement and inclusivity**, addressing a major gap in current educational technologies for sensory-challenged learners.

Researchers have explored a variety of approaches to support inclusive learning for children with sensory challenges. Several studies have focused on sign language recognition and translation systems, while others have explored the role of emotionally intelligent avatars and assistive technologies. However, few solutions integrate both accessibility and emotional expressiveness.

One study investigated the development of automatic sign language translation systems that convert text or speech into animated sign gestures. While these systems improved accessibility, they were limited in their ability to capture the emotional nuances of communication, which are essential in sign language expression. [1]

Another research initiative explored the use of avatar-based sign language interpreters for digital learning platforms. These avatars were capable of delivering sign gestures but often lacked realistic facial expressions and emotional cues, making the interaction less engaging for children. [2]

Studies on emotion recognition in education have demonstrated that learners benefit significantly when emotional context is integrated into digital tools. For example, systems that used sentiment analysis to detect tone in text improved comprehension and engagement among children. However, these systems were not directly applied to sign language translation. [3]

Research on assistive technologies for deaf education has also highlighted the importance of combining visual, expressive, and interactive elements. One project developed an interactive storytelling platform for deaf children, using visual animations to enhance comprehension. While successful in improving learning outcomes, it lacked direct text-to-sign translation and emotional depth. [4]

More recent approaches have begun experimenting with AI-driven expressive avatars that use facial expressions and gestures to simulate emotions. These systems demonstrated potential for creating more natural communication experiences but have yet to be applied specifically to text-to-sign translation for educational use. [5]

The gap in current literature lies in the integration of text-to-sign language translation with emotional expressiveness, designed specifically for children with sensory challenges. Most existing systems either prioritize accessibility (literal translation) or emotional intelligence (avatars with expressions), but not both together in a unified learning tool.

This research seeks to address that gap by developing a system that not only provides sign language translation but also incorporates emotional tones and facial cues, creating a richer and more inclusive educational experience. By doing so, it will advance the field of assistive learning technologies and contribute to the broader vision of inclusive education.

1.2 RESEARCH GAP

Identifying research gaps is crucial for advancing inclusive education and ensuring that new studies address the unmet needs of children with sensory challenges. While significant progress has been made in assistive technologies, existing research has not fully explored the integration of **sign language translation with emotional expressiveness**. This leaves a major opportunity for innovation, as emotional cues are vital for improving comprehension, engagement, and overall communication effectiveness. A review of existing literature highlights several areas where this study, the development of an **Emotionally Expressive Reader**, aims to contribute.

The first research paper [1] discusses automatic text-to-sign language translation systems that convert spoken or written words into sign gestures. While these systems improve accessibility, they are limited to literal translations and often lack **emotional depth**, which is essential in sign language communication. Emotional cues such as facial expressions and body posture are often excluded, reducing the effectiveness of communication. Therefore, there is a gap in creating systems that combine both **linguistic accuracy and emotional expressiveness**.

The second research initiative [2] explored avatar-based sign language interpreters for digital learning platforms. Although these avatars enhanced accessibility, they frequently failed to incorporate **realistic facial expressions and emotional tones**, making the interaction less engaging and less natural. This indicates the need for research that integrates **emotion recognition and expressive avatars** into text-to-sign systems.

The third research study [3] focused on emotion recognition in educational technologies, showing that learners benefited when digital tools conveyed emotional context. However, these systems primarily targeted **text-to-speech applications**, not **text-to-sign translation**. This highlights a gap in applying emotion detection techniques directly to assistive tools for deaf and hard-of-hearing learners.

Another relevant study [4] investigated interactive storytelling platforms for deaf children using visual animations. While the project improved comprehension and engagement, it lacked **direct sign language integration** and did not address the emotional depth of communication. Therefore, further research is required to design systems that combine **interactive learning, sign language, and emotional cues**.

Recent studies [5] have experimented with expressive avatars capable of simulating facial emotions. While these show promise for more natural communication, their application in **educational settings**

for sensory-challenged children is still limited. Additionally, few attempts have been made to integrate such avatars into real-time text-to-sign translation tools designed specifically for children.

In conclusion, this research identifies a clear gap in existing literature: while studies have addressed either **sign language translation** or **emotional expressiveness**, few have integrated both into a single, child-friendly educational tool. The proposed **Emotionally Expressive Reader** seeks to fill this gap by combining **text-to-sign translation, emotion recognition, and expressive avatars**, creating a system that not only improves accessibility but also enhances emotional engagement and comprehension. By doing so, this project will advance inclusive education technologies and contribute meaningfully to the development of innovative learning tools for children with sensory challenges.

Research Gap Feature	Research 1 [9]	Research 2 [3]	Research 3 [4]	Research 4 [10]	Proposed Research Solution
Integration of Emotional Expressiveness in Sign Language Translation	✓	✗	✗	✗	✓
Real-Time Text-to-Sign Conversion with Emotion Recognition	✗	✗	✗	✗	✓
Child-Centered Assistive Learning Tools Combining Accessibility and Engagement	✗	✓	✗	✓	✓

Table 1: Research Gap for component

1.3 RESEARCH PROBLEM

The current landscape of assistive learning tools for children with sensory challenges, particularly those who are deaf or hard of hearing, is marked by several limitations that undermine their effectiveness and inclusivity. One of the most pressing problems is the lack of **emotional expressiveness** in existing text-to-sign language systems. Most available solutions focus solely on providing literal translations of text into sign gestures, but they fail to capture the **facial expressions, body language, and emotional tone** that are integral components of natural sign language communication. [1] The absence of these features often leads to reduced comprehension, disengagement, and a diminished ability for children to fully grasp the context and meaning of the content.

In addition to this, the majority of current systems rely on **static or robotic avatars** that lack realism and adaptability. These avatars often deliver signs in a mechanical manner, which does not reflect the dynamic and expressive nature of real human communication. [2] Such limitations make it difficult for children to build a strong emotional connection with the learning material, which is a key factor in sustaining engagement and improving learning outcomes.

Another critical issue lies in the **limited accessibility and customization** of existing tools. Many assistive systems are designed for general use and do not address the specific needs of children with sensory challenges. They often lack features such as adaptive difficulty levels, culturally relevant sign language variations, or the integration of emotional cues tailored to young learners. [3] This one-size-fits-all approach hinders inclusivity and reduces the overall educational impact.

Furthermore, there is a **gap in real-time emotion-aware processing**. While some educational technologies have explored sentiment detection for text-to-speech applications, very few have extended this capability to **text-to-sign translation with expressive emotional feedback**. [4] This lack of real-time emotional adaptation restricts the ability of children to experience natural, interactive, and engaging communication during reading activities.

These challenges highlight the urgent need for a more **emotionally expressive, child-centered assistive reader** that integrates sign language translation with emotional cues. By combining **AI-driven emotion recognition, expressive avatars, and sign language animation**, such a system can provide a richer, more engaging, and inclusive learning experience. The proposed **Emotionally Expressive Reader** aims to fill this gap by enabling children with sensory challenges to not only access information but also connect with its emotional meaning—ultimately supporting better comprehension, engagement, and educational outcomes.

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to develop an **Emotionally Expressive Reader** that translates written text into sign language with integrated **emotional expressiveness** (facial expressions, tone, and gestures). The system aims to improve **comprehension, engagement, and inclusivity** for children with sensory challenges, particularly those who are deaf or hard of hearing.

2.2 Specific Objectives

1. Develop a Text-to-Sign Translation Engine

This objective focuses on creating a module that accurately converts written text into sign language animations. It is **Specific** to providing linguistic accessibility, **Measurable** by assessing translation accuracy and clarity, **Achievable** using existing natural language processing and sign language datasets, **Realistic** as it builds upon existing sign language recognition technologies, and **Time-bound** to be developed in the early stages of the project.

2. Integrate Emotion Recognition for Expressive Translation

The goal is to design an emotion recognition system that detects the emotional tone of the text (e.g., happy, sad, surprised) and reflects it through expressive avatars. This objective is **Specific** to enhancing emotional engagement, **Measurable** by evaluating user comprehension and emotional recognition rates, **Achievable** using AI-based sentiment analysis and avatar design, **Realistic** with current advances in emotion AI, and **Time-bound** to align with prototype development milestones

3. Design an Expressive Avatar-Based Reader for Children

The objective is to create a child-friendly, animated avatar capable of delivering sign language with facial expressions and gestures. It is **Specific** to supporting learning for children with sensory

challenges, **Measurable** by user satisfaction and engagement levels during testing, **Achievable** through modern 2D/3D animation tools, **Realistic** with simplified avatar models, and **Time-bound** within the app development cycle.

4. Implement an Interactive and Inclusive Reading Platform

This objective focuses on developing a user-friendly reading platform that integrates the translation engine and expressive avatars into an interactive environment for children. It is **Specific** to improving accessibility and inclusivity, **Measurable** by ease-of-use surveys and interaction success rates, **Achievable** through mobile or tablet-based applications, **Realistic** with current software development frameworks, and **Time-bound** to coincide with the final stage of the project.

3. METHODOLOGY

This research aims to develop an **Emotionally Expressive Reader** that translates written text into sign language supported by emotional expressions. The methodology follows a structured approach, integrating natural language processing (NLP), sentiment analysis, sign language animation, and mobile/desktop application development. Each task and sub-task within the methodology is designed to ensure that the system operates **accurately, expressively, and in a child-friendly manner**.

3.1 Key Pillars and Application of Technologies

1. Natural Language Processing (NLP):

NLP forms the foundation of the system, enabling the analysis of input text to extract linguistic meaning and structure. This ensures that sentences are correctly segmented and mapped into corresponding sign language equivalents. By leveraging NLP, the system will maintain semantic accuracy and improve sign representation for deaf children.

2. Emotion and Sentiment Analysis:

Emotion detection techniques (using AI-based sentiment analysis and tone detection) will be applied to identify the emotional context of the text. Recognized emotions (such as happiness, sadness, anger, or surprise) will then be reflected in the avatar's facial expressions and gestures to provide a richer and more engaging reading experience.

3. Avatar-Based Sign Language Animation:

An expressive 2D/3D avatar will serve as the visual interpreter, performing sign language with synchronized facial expressions and gestures. The avatar will be designed to be child-friendly and engaging, ensuring accessibility and comfort for children with sensory challenges.

4. Interactive Application Development:

A mobile/tablet application will be developed as the main interface for children. The app will allow users to input or select text, view translations in sign language, and experience expressive storytelling through the animated avatar. Features like replay, speed adjustment, and visual aids will enhance interactivity.

3.2 Proposed Approach and Task Breakdown

1. Development of Text-to-Sign Translation Engine

The first major task is to create a robust engine that converts written text into sign language sequences.

- **Sub-task 1: NLP Model Integration** – Apply NLP techniques to process input text and map it to corresponding sign language glosses.
- **Sub-task 2: Sign Language Dataset Training** – Train the model with an existing sign language dataset (e.g., ASL/ISL/BSL depending on scope).
- **Sub-task 3: Accuracy Testing** – Validate translations for correctness and fluency through expert review and user testing.

2. Implementation of Emotionally Expressive Features

The second task focuses on enhancing expressiveness through emotion recognition and avatar design.

- **Sub-task 1: Sentiment Analysis Module** – Integrate an AI model to analyze the emotional tone of input text.
- **Sub-task 2: Avatar Expression Mapping** – Map detected emotions to facial expressions, body gestures, and signing style of the avatar.
- **Sub-task 3: User Validation** – Conduct testing with children and educators to ensure emotional clarity and effectiveness.

3. Design and Development of Expressive Avatar

This task involves building a child-friendly avatar capable of delivering sign language with accurate gestures and emotions.

- **Sub-task 1: Avatar Design and Animation** – Create a visually appealing and accessible avatar.
- **Sub-task 2: Gesture Synchronization** – Ensure synchronization between signs, expressions, and spoken/written text.
- **Sub-task 3: Prototype Evaluation** – Test the avatar's performance with small groups of children.

4. Creation of an Interactive Reading Platform

The final task is to integrate the translation engine and avatar into an interactive platform.

- **Sub-task 1: Application Interface Development** – Develop a mobile/tablet app with simple navigation.
- **Sub-task 2: Accessibility Features** – Include features such as adjustable playback speed, subtitles, and replay options.
- **Sub-task 3: User Testing and Feedback** – Evaluate usability, engagement, and comprehension levels among children with sensory challenges.

3.1 System Diagram

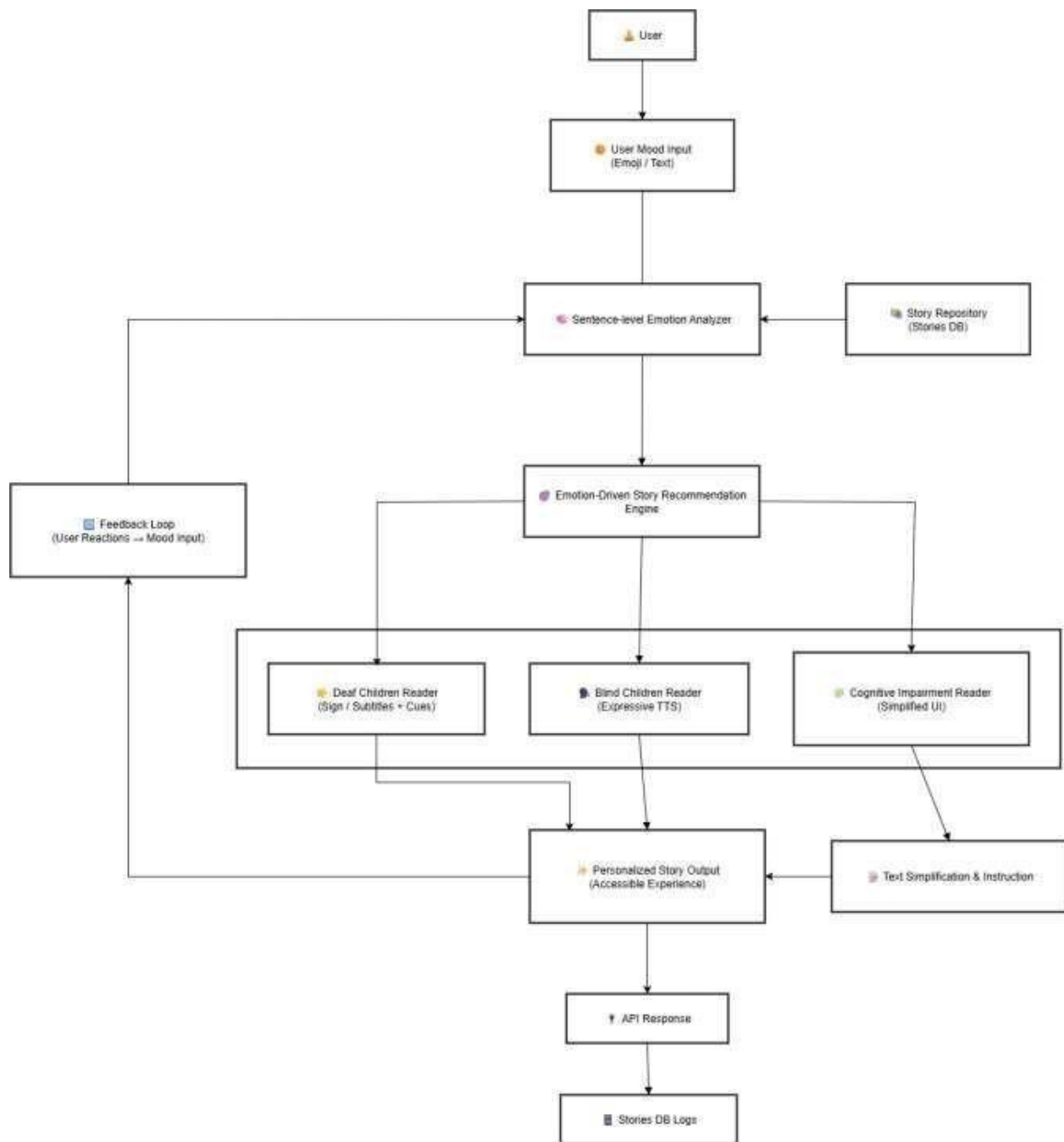


Figure 1: Component Diagram

3.2 Proposed Project Plan

3.2.1 Data Collection

The data required for this project includes text datasets, sign language glosses, and emotional expression annotations. These datasets will serve as the foundation for training the text-to-sign language translation model and integrating emotional expressiveness into the sign animations. The data will be preprocessed and stored securely to ensure accuracy, consistency, and ethical use.

1. Text Data:

- **Data Needed:** Written sentences, short stories, and educational texts that are commonly used for children's learning and reading activities.
- **Collection Method:** Publicly available open-source text corpora for children's literature and educational materials will be collected. Additional sample stories will be prepared by the research team to ensure relevance to the learning context.

2. Sign Language Data:

- **Data Needed:** Sign language glosses (text-based symbolic representations of signs), video recordings of sign language gestures, and datasets linking text with corresponding signs.
- **Collection Method:** Existing open-source sign language datasets (e.g., RWTH-PHOENIX-Weather, ASLLVD, or regional equivalents) will be used. Supplementary data may be collected through collaboration with sign language interpreters to ensure contextual accuracy and child-friendly sign selection.

3. Emotional Expression Data:

- **Data Needed:** Annotated datasets linking sentences with emotional tones (e.g., happy, sad, excited, neutral) and corresponding non-manual features such as facial expressions, body posture, and intensity levels.

- **Collection Method:** Sentiment analysis datasets will be integrated with sign language avatar training data. Additional emotional labeling will be carried out by experts or volunteers to enrich the dataset with expressive variations suitable for children's comprehension.

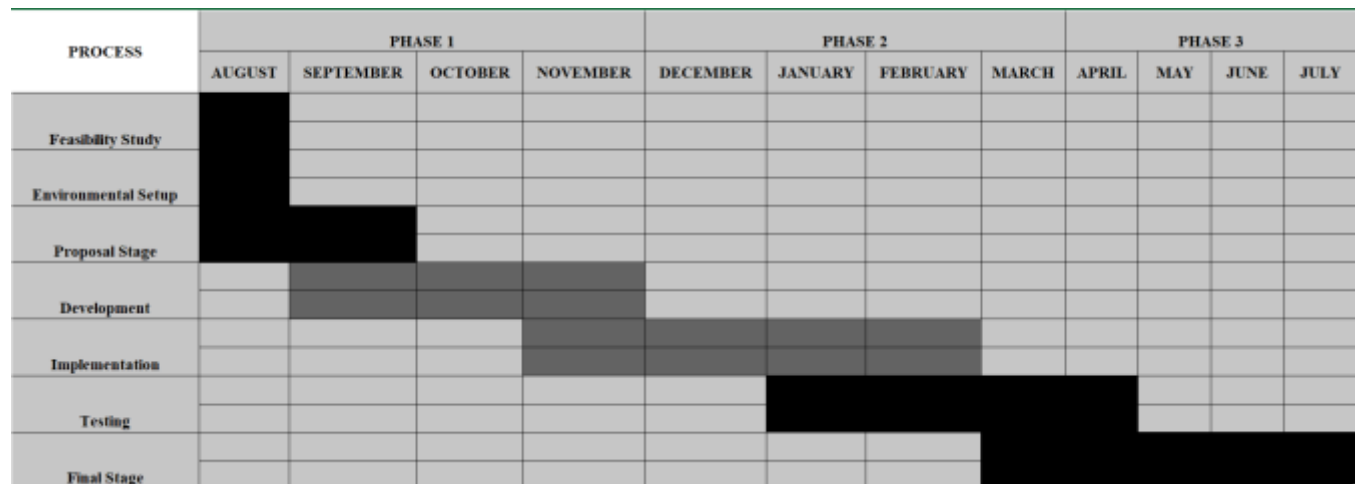


Figure 2: Gantt Chart

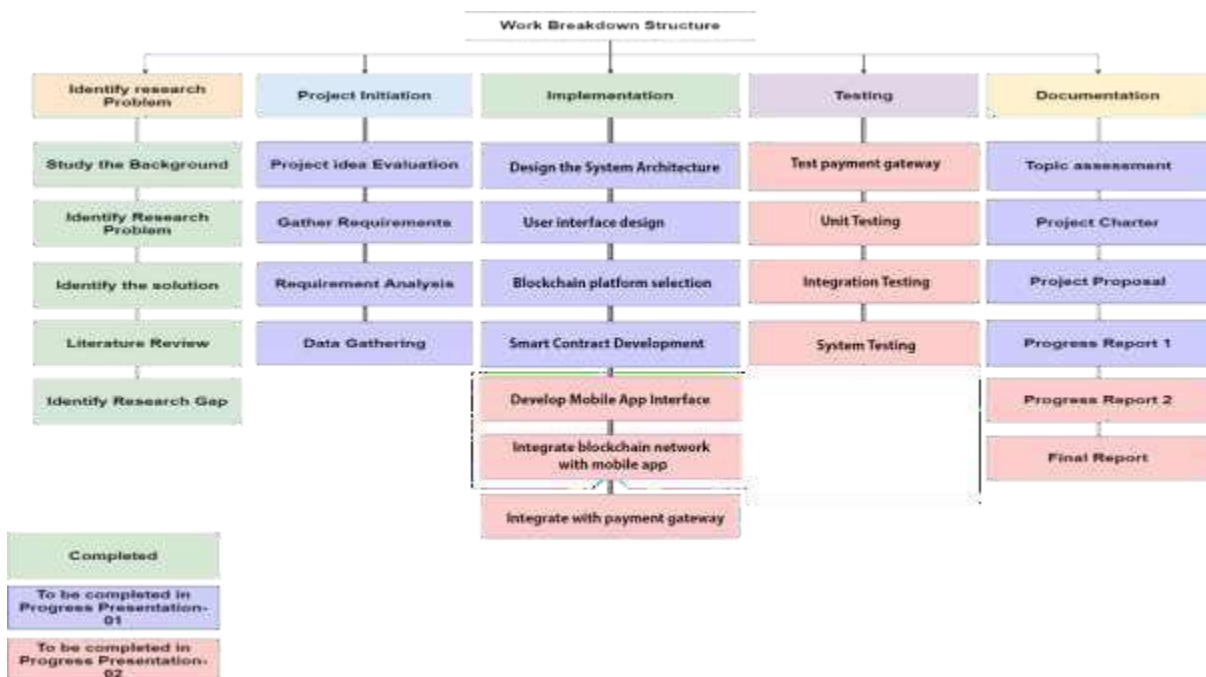


Figure 3: Work Breakdown Structure

3.3 Anticipated Conclusion

The anticipated conclusion of this project is the successful development and deployment of a text-to-sign language translation system enriched with emotional expressiveness, specifically designed to support children with sensory challenges. The system will significantly improve accessibility to written text, providing a child-friendly platform where stories, educational content, and daily communication can be understood in sign language with natural emotional expressions.

Results:

- **Accuracy and Expressiveness:** By leveraging natural language processing (NLP) and sign language datasets, the system will ensure accurate translation of text into sign language. The addition of emotional expressiveness (facial expressions, tone, and body language in avatars) will make the output more engaging and easier for children to understand.
- **Improved Learning Experience:** The mobile/desktop application will provide an intuitive and interactive interface that allows children to visualize text in sign language. This will encourage active learning, enhance comprehension, and build confidence in reading for children with hearing or sensory challenges.
- **Enhanced Accessibility:** The system will serve as an inclusive tool, reducing communication barriers and promoting equal opportunities in education by ensuring that children who rely on sign language have access to expressive and meaningful text translations.

Application:

- **Scalability:** The text-to-sign language system is scalable and can be expanded to support multiple sign languages, educational curricula, and age-specific content. It can also be integrated into e-learning platforms, digital libraries, and assistive technologies to broaden its impact.
- **Inclusivity in Education:** Schools, parents, and caregivers will be able to adopt this system as part of inclusive education strategies, ensuring that children with sensory challenges are not left behind in their learning journey.

Real-World Use:

- **Educational Institutions and Special Needs Schools:** The system is designed for direct implementation in learning environments, helping teachers and caregivers communicate stories and lessons effectively to children with hearing or communication difficulties.
- **Enhanced Child Engagement:** Children will benefit from emotionally expressive sign translations that bring stories and lessons to life, making learning both enjoyable and effective.

This will improve comprehension, retention, and emotional connection to the content.

In conclusion, this project will provide an innovative solution for inclusive education, addressing current accessibility challenges with advanced text-to-sign language translation technology. It paves the way for a more inclusive digital future, ensuring that children with sensory challenges can learn, engage, and thrive equally in educational environments.

3.4 Commercialization

The commercialization strategy for the “**Emotionally Expressive Text-to-Sign Language Reader**” is designed to target key markets such as special education schools, inclusive learning institutions, therapy centers, non-profit organizations supporting children with hearing impairments, and government-funded educational initiatives. These sectors represent a growing demand for innovative assistive technologies that can enhance accessibility, improve learning outcomes, and support inclusive education.

For educational institutions and therapy centers, the system presents considerable advantages by minimizing the reliance on manual interpretation, thereby reducing operational challenges and costs. The automated text-to-sign language translation, enriched with emotional expressiveness, ensures that learning materials are delivered in an engaging, accurate, and child-friendly manner. This improves student comprehension while also reducing the workload on educators and interpreters.

Economically, the system provides value by offering cost-effective accessibility solutions for schools and organizations that may not have access to full-time sign language interpreters. Its modular and scalable design allows integration with e-learning platforms, mobile applications, and digital libraries, ensuring long-term adaptability and relevance as technology evolves.

By focusing on these target markets and emphasizing the dual benefits for both learners and educators, the **Text-to-Sign Language Reader** is well-positioned to become a leading solution in the assistive technology market. Its unique capability to combine linguistic accuracy with emotional expressiveness provides a strong differentiator, making it suitable for widespread adoption across educational and therapeutic contexts.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- **Text-to-Sign Language Conversion:**

The system must accurately convert input text into corresponding sign language animations. Each word, phrase, or sentence should be translated into standardized sign gestures to ensure effective communication for deaf and hard-of-hearing users.

- **Emotionally Expressive Sign Rendering:**

The system must integrate emotional expressions (e.g., happy, sad, excited, neutral) into the sign animations. This ensures that the meaning and tone of the message are conveyed effectively, supporting better comprehension for children with sensory challenges.

- **Real-Time Translation:**

The application must provide real-time text-to-sign translation. As a user types or inputs text, the system should immediately generate and display the corresponding sign language animation without significant delay.

- **User-Friendly Interface for Children:**

The system must include an intuitive and engaging interface designed for children. This includes features such as clear visuals, simple navigation, and interactive elements to enhance the learning experience.

- **Accessibility Features:**

The system must include additional accessibility options such as text resizing, customizable color themes, and playback controls for sign animations (pause, replay, speed adjustment) to cater to children with different sensory or learning needs.

- **Feedback and Confirmation Mechanism:**

After each translation, the system must provide a confirmation output (e.g., visual cues, playback summary, or voice/text feedback) so that users and educators can verify the accuracy of the generated signs.

4.2 User Requirements

- **Ease of Use:**

The mobile app should be intuitive and easy to navigate, with a clean and responsive user interface that accommodates users of all technical levels.

- **Security and Privacy:**

Users expect that their personal and financial information will be protected.

- **Accessibility:**

The app should be accessible to users with disabilities, incorporating features like voice commands and screen reader compatibility.

4.3 System Requirements

Hardware Requirements:

- **Servers or Cloud Infrastructure:** To host the text-to-sign translation engine, manage the animation library, and handle real-time processing.
- **User Devices:** Mobile devices, tablets, or desktop computers for running the application and displaying sign language animations.
- **Peripheral Devices (Optional):** Cameras and microphones for possible future integration of speech-to-text and gesture recognition features.

Software Requirements:

- **Sign Language Translation Engine:** Core software module responsible for converting text input into corresponding sign language gestures.
- **Animation Rendering Framework:** Tools for displaying 3D/2D animated avatars capable of performing expressive sign language movements.

- **Mobile and Web Application Development Frameworks:** For building cross-platform apps (e.g., Flutter, React Native) to ensure accessibility on multiple devices.
- **Database System:** To securely store sign language datasets, user preferences, and learning progress data.
- **Accessibility Support Libraries:** To integrate features such as text resizing, color adjustments, and interactive controls for children with sensory challenges.

Network Requirements:

- **Stable Internet Connection:** For cloud-based translation, data synchronization, and access to large sign language datasets.
- **Offline Mode (Optional):** Support for offline access with locally cached sign animations for areas with limited connectivity.
- **Cross-Platform Compatibility:** Ensure smooth communication between servers, databases, and mobile/web applications across different operating systems and network environments.

4.4 Non-Functional Requirements

- **Performance:**

The system should process text input and display the corresponding sign language animation within a few seconds to ensure real-time communication and learning support.

- **Scalability:**

The system should be capable of supporting an increasing number of users, multiple devices, and expanding sign language datasets without performance degradation.

- **Security:**

The system must ensure the protection of user data, especially children's information, through encryption and secure authentication mechanisms, preventing unauthorized access or misuse.

- **Reliability:**

The system must consistently deliver accurate text-to-sign translations and maintain high availability, minimizing downtime to ensure uninterrupted learning and communication.

5. BUDGET AND BUDGET JUSTIFICATION

Component	Amount (LKR)
Web Hosting	9,600
Development Tools and Techniques	22,400
Cloud Hosting and Data Storage	12,000
Paper publishing cost	50,000
Total	94,000

Table 2: Budget Distribution

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